100 SLO-Based Short Questions

CLASS 11 • SUMMATIVE THEORY • 3 MARKS PER QUESTION

Each question includes its SLO code and a suggested split of the 3 marks. In numericals, show the formula, substitution and unit. SLO codes follow the supplied assessment framework, including its P-09-E and P-12-C prefixes.

Measurements and mechanics

Q01–Q06 • 6 QUESTIONS

01 **P-11-A-02** 1+2 marks

Pressure is force per unit area. Write the SI unit of force in base units, then express the pascal entirely in SI base units.

02 **P-11-A-03** 1+2 marks

A student writes $s = ut + \frac{1}{2}at$, where s is displacement, u is initial speed, a is acceleration and t is time. Compare the dimensions of the two terms on the right and correct the power of t in the second term.

03 **P-11-A-05** 1+1+1 marks

A reference mass is 50.0 g. Balance A reads 49.9, 50.0 and 50.1 g. Balance B reads 52.0, 52.0 and 52.1 g. Identify which balance is more precise and which is more accurate. Use the readings to justify your choices.

04 **P-11-A-06** 1+2 marks

A rectangular card measures (20.0 ± 0.2) cm by (10.0 ± 0.1) cm. Calculate its area and the absolute uncertainty in the area, using the percentage-uncertainty rule.

05 **P-11-B-01** 2+1 marks

A cable pulls a trolley with a force of 80 N at 30° above the horizontal. Calculate the horizontal and vertical components. Which component reduces the normal force on the trolley?

06 **P-11-B-05** 2+1 marks

A ball rolls off a 20 m high ledge at 6.0 m/s horizontally. Calculate the time to reach the ground and the horizontal distance from the foot of the ledge. Use $g = 10$ m/s² and neglect air resistance.

Topic guide

07–17 Rotational and circular motion

18–24 Work and kinetic energy

25–33 Fluid mechanics

34–40 Deformation of solids

41–51 Heat and thermodynamics

52–62 Waves

63–69 Electrostatics

70–78 Current electricity

79–87 Magnetism and electromagnetic induction

88–92 Special relativity

93–100 Nuclear and particle physics

Rotational and circular motion

Q07–Q17 • 11 QUESTIONS

07 P-11-B-07, B-08 1+2 marks

A bicycle wheel turns through 150° in 0.50 s. Convert its angular displacement to radians and find its average angular speed in rad/s.

08 P-11-B-09 1+2 marks

A fan slows uniformly from 24 rad/s to 8.0 rad/s in 4.0 s. Calculate its angular acceleration and angular displacement during this interval.

09 P-11-B-08 2+1 marks

Two dots on a rigid rotating disc are 4.0 cm and 12 cm from its axis. The disc rotates at 15 rad/s. Find each dot’s linear speed and explain why their angular speeds are equal.

10 P-11-B-10 2+1 marks

A bead moves at constant speed around a circular track. Explain how it can accelerate at constant speed. If the track ceases to exert a force and no other horizontal force acts, state the bead’s immediate direction of motion.

11 P-11-B-14, B-15 1+2 marks

A 900 kg car takes a flat bend of radius 45 m at 12 m/s. Identify the force that provides the centripetal force and calculate its required magnitude.

12 P-11-B-16 1+2 marks

An astronaut lets go of a pen inside an orbiting spacecraft. Describe what the astronaut sees and explain the observation in terms of the motion of the pen and the spacecraft under gravity.

13 P-11-B-17 1+2 marks

A rotating space station has a circular living area along its rim. Which way is “down” for the crew? Explain how the floor’s normal force produces apparent weight.

14 P-11-B-18 2+1 marks

Two 0.50 kg masses are attached to a light rod at distances of 0.20 m and 0.40 m from its fixed axis. Find their total moment of inertia and their angular momentum when they rotate at 6.0 rad/s.

15 P-11-B-19 1+2 marks

A disc with a moment of inertia of 0.40 kg·m² is driven by a torque of 1.8 N·m. A frictional torque of 0.20 N·m opposes the rotation. Find the net torque and the angular acceleration.

16 P-11-B-20 1+2 marks

A person on a freely rotating stool draws two handheld masses inward. Explain the change in angular speed. If the total moment of inertia falls to three quarters of its initial value, find the ratio of final to initial angular speed.

17 P-11-B-21 1+2 marks

In a spinning centrifuge tube containing a suspension, where do denser particles tend to collect? Explain how rotation helps separate them from the less dense liquid.

Work and kinetic energy

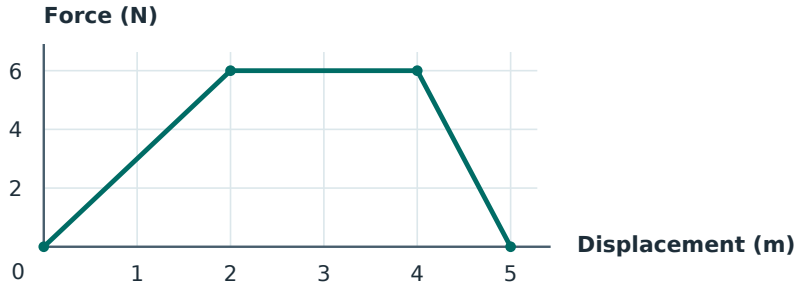
Q18–Q24 • 7 QUESTIONS

18 P-11-B-29 1+2 marks

A cart starts from rest and is pushed by a constant net force F through a distance s. Write the work done and use $v^2 = 2as$ to express it in terms of the cart's mass and final speed.

19 P-11-B-30 1+2 marks

Use the force–displacement graph below to find the work done from 0 to 2 m and the total work done from 0 to 5 m.



20 P-11-B-31 1+2 marks

A box is lifted and then lowered back to its starting height. A second box is pushed around a closed path on a rough floor. Compare the net work by gravity in the first case with that by friction in the second. Explain the difference.

21 P-11-B-32 1+2 marks

A 4.0 kg crate starts from rest. A horizontal pull of 20 N moves it 5.0 m across a floor against an 8.0 N friction force. Find the net work and the final speed.

22 P-11-B-32 1+2 marks

A 60 kg cyclist slows from 8.0 m/s to 4.0 m/s while coasting 12 m along a level road. Treat 60 kg as the total moving mass. Find the change in kinetic energy and the magnitude of the average resisting force.

23 P-11-B-12 2+1 marks

On a straight track, the first cart moves right at 5.0 m/s and the second moves left at 2.0 m/s. After an elastic collision, the first moves left at 1.0 m/s and the second moves right. Use relative speeds to find the second cart's final speed and state what happens to their total kinetic energy.

24 P-11-B-13 1+2 marks

Two identical trolleys approach each other at equal speeds and stick together. Predict their final motion. Explain how their total momentum can remain constant while their total kinetic energy changes.

Fluid mechanics

Q25–Q33 • 9 QUESTIONS

25 P-11-B-33 1+2 marks

A block floats with 75% of its volume below the surface of fresh water. Write the force balance and calculate the density of the block. Take the density of water as 1000 kg/m³.

26 P-11-B-34 2+1 marks

A steel ship floats in the sea, while a solid piece of the same steel sinks. Explain the role of the hollow hull and what happens to the ship’s submerged volume when cargo is added.

27 P-11-B-35, B-41 1+2 marks

An ideal fluid is assumed to be incompressible and non-viscous. Explain these two assumptions and identify which one fails when neighbouring layers of real oil exert a drag on each other.

28 P-11-B-36 1+2 marks

Water flows steadily through a pipe at 0.60 m/s where its cross-sectional area is 12 cm². Find the volume flow rate in cm³/s and the speed in a section of area 4.0 cm².

29 P-11-B-37, B-38 1+2 marks

The outlet area of a hose is reduced to one third of its original value while the volume flow rate is kept constant. Predict the change in outlet speed and justify it using conservation of mass.

30 P-11-B-40 1+2 marks

Treat water as an ideal fluid moving steadily upward through a pipe of constant cross-section. An upper point is 2.0 m above a lower point. Explain why the speeds are equal and use Bernoulli’s equation to find the pressure drop. Take ρ = 1000 kg/m³ and g = 10 m/s².

31 P-11-B-41, B-42 1+2 marks

Identical beads are released in water and glycerine at the same temperature. Glycerine is much more viscous. At the same bead speed, which liquid exerts more viscous drag? Explain how the moving liquid layers resist the bead’s motion.

32 P-11-B-43 1+2 marks

A vortex in an ordinary liquid gradually dies away after stirring stops. Explain why. Predict how a vortex in an idealized superfluid would behave and relate your answer to its viscosity.

33 P-11-B-44 1+2 marks

A Venturi filter pump has a side tube at the narrow throat of its main passage. Explain why placing the side tube at the throat helps draw air in when water flows through the pump.

Deformation of solids

Q34–Q40 • 7 QUESTIONS

34 P-11-B-22 1+1+1 marks

One sample has a repeating three-dimensional particle arrangement. Another is made of long molecular chains. Identify the characteristic structure in each case and state whether an ordinary glass has the same long-range order as the first sample.

35 P-11-B-23, B-24 1+2 marks

A straight rod is pushed inward at both ends and shortens by 0.20 mm. Identify the type of deformation and find the magnitude of its strain if its original length is 0.50 m. State the unit, if any, of strain.

36 P-11-B-24 1+1+1 marks

A 2.0 m wire of cross-sectional area 1.0 mm^2 extends by 1.0 mm under a tensile force of 50 N. Calculate the tensile stress, tensile strain and Young modulus.

37 P-11-B-24 1+2 marks

Two wires of the same metal carry the same load and have equal cross-sectional areas. Wire A is twice as long as wire B. Compare their strains and extensions, assuming both are within the proportional limit.

38 P-11-B-26 1+2 marks

A wire is stretched by 3.0 mm. After unloading, its length is 0.80 mm greater than its original length. Find how much extension was recovered and explain what the permanent extension reveals about the elastic limit.

39 P-11-B-27, B-28 1+2 marks

Within the proportional limit, a spring needs 18 N to hold it at an extension of 6.0 cm. Sketch the force–extension graph up to that point and use its area to find the stored elastic energy.

40 P-11-B-28 1+2 marks

A spring stores 0.20 J at an extension of 2.0 cm. What energy will it store at 6.0 cm if it still obeys Hooke’s law? Justify using the relationship between elastic energy and extension.

Heat and thermodynamics

Q41–Q51 • 11 QUESTIONS

41 P-12-C-01, P-11-C-02 1+2 marks

Two metal blocks have different masses but are at the same temperature. Predict the net heat transfer if they touch. If one block is then heated without a change of state, explain what happens to its internal energy.

42 P-11-C-03, C-04 1+2 marks

A rigid 0.020 m^3 vessel holds 0.50 mol of an ideal gas at 300 K. Calculate the pressure. Write the equivalent equation using the number of molecules N and state how k is related to R. Use $R = 8.31$ J/(mol·K).

43 P-11-C-05 1+2 marks

Under what pressure and temperature conditions does a real gas behave most like an ideal gas? Explain why the molecular volume and intermolecular attractions can often be neglected under those conditions.

44 P-11-C-06, C-08 1+2 marks

A gas expands from 0.010 m^3 to 0.014 m^3 at 1.0×10^5 Pa and absorbs 550 J of heat. Calculate the work done by the gas and its change in internal energy.

45 P-11-C-07, C-08 2+1 marks

A gas loses 120 J of heat while a piston does 200 J of work on it. State the sign of each transfer using $Q = \Delta U + W$, where W is work done by the gas. Calculate ΔU.

46 P-11-C-09 1+2 marks

An ideal gas in a sealed rigid container is heated. Predict the change in its pressure and explain it using molecular speeds and collisions with the walls.

47 P-11-C-10 1+2 marks

An ideal gas has a pressure of 120 kPa at a volume of 2.0 L. It is compressed isothermally to 0.80 L. Find the final pressure and sketch the pressure–volume curve, marking both states.

48 P-11-C-14, C-15 1+2 marks

A hot metal block is dropped into cooler water in an insulated container. Predict the direction of heat transfer. Explain why the process is irreversible and relate your answer to the second law.

49 P-11-C-13, C-15 1+2 marks

An engine takes in 800 J of heat and delivers 300 J of work per cycle. How much heat does it reject? Explain why a cyclic heat engine cannot convert all the heat taken from a single reservoir into work.

50 P-11-C-23 2+1 marks

A proposed heat engine works between reservoirs at 600 K and 300 K. Calculate the Carnot efficiency and decide whether a claimed efficiency of 60% is possible.

51 P-11-C-19, C-20 1+2 marks

A spinning flywheel slows to rest because of friction in its bearings. State where its mechanical energy goes and explain how this change illustrates energy degradation and entropy increase.

Waves

Q52–Q62 • 11 QUESTIONS

52 P-11-D-01 1+2 marks

A progressive wave carries 6.0 W through an area of 0.50 m². Find its intensity. If its amplitude is then halved under otherwise identical conditions, find its new intensity.

53 P-11-D-02, D-03 1+2 marks

A vehicle’s siren emits 680 Hz as it moves away from a stationary listener at 20 m/s. State whether the listener hears a higher or lower frequency and calculate it. Take the speed of sound as 340 m/s.

54 P-11-D-04 1+2 marks

A stationary weather radar sends waves toward raindrops moving directly toward it. Predict whether the echo frequency increases or decreases and explain how the size of the shift helps estimate the drops’ speed.

55 P-11-D-05 1+2 marks

A wave on a rope can have either vertical or horizontal oscillations. A vertical slit allows only the vertical oscillations through. Identify the phenomenon and explain why a longitudinal sound wave in air cannot show the same effect.

56 P-11-D-06 2+1 marks

Plane-polarized light of intensity 80 W/m² passes through an analyser whose axis is 30° to the initial polarization. Calculate the transmitted intensity. What happens if a second analyser is placed at 90° to the first analyser?

57 P-11-D-07, D-08 1+2 marks

Two equal-amplitude coherent sources start in phase. At point X their waves travel paths differing by 1½ wavelengths. At point Y the difference is 2 wavelengths. Identify the interference at each point and justify your choices using the path differences.

58 P-11-D-09 1+2 marks

At a listener’s ear, a headphone creates a sound wave with the same frequency and amplitude as an unwanted hum. What phase difference should it have? Explain why the hum is reduced at the ear.

59 P-11-D-11, D-12 2+1 marks

Sketch the third harmonic on a string fixed at both ends, labelling nodes and antinodes. Write the wavelength in terms of the string length L.

60 P-11-D-13 1+2 marks

Straight water waves meet a barrier with an opening. Compare the spreading when the opening is about one wavelength wide and when it is ten wavelengths wide. Sketch the wavefronts beyond both openings.

61 P-11-D-14 1+2 marks

Two independent lamps emit light of the same colour. State the phase condition required for coherence and explain why splitting the light from a single source can produce a steadier interference pattern.

62 P-11-D-15, D-16 1+2 marks

A violin string and a 440 Hz reference tone produce 3 beats per second. Give the two possible string frequencies. Tightening the string raises its frequency by 1 Hz, and the beat rate increases. Which frequency was the string originally producing? Explain.

Electrostatics

Q63–Q69 • 7 QUESTIONS

63 P-11-E-01, E-02 2+1 marks

A +3.0 µC test charge experiences a force of 0.012 N east at a point. Find the magnitude and direction of the field. What direction of force would a negative charge experience there?

64 P-11-E-02, E-04 2+1 marks

Two parallel plates 4.0 mm apart have a potential difference of 24 V. Find the magnitude of the uniform electric field. An electron is released from rest between the plates. State which plate it accelerates toward and why.

65 P-11-E-03 2+1 marks

Sketch the electric field lines for two equal positive point charges separated by a small distance. Mark the directions and explain why the field at their midpoint is zero.

66 P-11-E-05, E-08 1+2 marks

An isolated conducting sphere of radius 0.10 m carries +4.0 nC. Explain how to model its charge for a point 0.30 m from the centre and calculate the field magnitude there. Use $k = 9.0 \times 10^9$ N·m²/C².

67 P-11-E-06 1+2 marks

A charged rod is brought near a closed, hollow, neutral metal box. Describe what happens to the free charges on the box and explain the resulting electric field inside the empty cavity once electrostatic equilibrium is reached.

68 P-11-E-07 2+1 marks

Two small charged spheres repel with a force of 0.18 N. Their separation is doubled and one charge is tripled. Find the new force and explain whether it remains repulsive.

69 P-11-E-08 1+2 marks

Charges +2.0 nC and −2.0 nC are 0.20 m apart. Determine the direction and magnitude of the net electric field at the midpoint. Use $k = 9.0 \times 10^9$ N·m²/C².

Current electricity

Q70–Q78 • 9 QUESTIONS

70 P-09-E-10 1+2 marks

A metal wire carries 2.0 A. Its cross-sectional area is doubled using the same metal, and the current is kept constant. State the relation between current and drift speed and find the ratio of new to original drift speed.

71 P-09-E-11, E-16 2+1 marks

A cell transfers 9.0 J of energy to 3.0 C of charge in the complete circuit. A lamp receives 6.0 J from the same charge. Calculate the cell’s e.m.f. and the p.d. across the lamp. Explain what the difference represents.

72 P-09-E-12 2+1 marks

Two heaters of resistances 4.0 Ω and 8.0 Ω are connected in turn across the same 12 V supply. Find the power of each and identify which transfers more energy in one minute.

73 P-09-E-13 1+2 marks

A wire of resistance 5.0 Ω is stretched uniformly until its length doubles. Its volume and resistivity remain constant. Find the new cross-sectional area as a fraction of the original and calculate the new resistance.

74 P-09-E-17, E-26 1+2 marks

A cell of e.m.f. 6.0 V and internal resistance 0.50 Ω delivers 2.0 A. Calculate its terminal p.d. and explain what happens to that p.d. when the external circuit is opened.

75 P-09-E-18, E-19 1+2 marks

At a junction, 4.0 A enters and 1.5 A leaves through one branch. Find the current in the only other branch, including its direction. State which quantity is conserved in the junction rule and which is conserved in the loop rule.

76 P-09-E-21, E-22 2+1 marks

A 6.0 Ω resistor and a 3.0 Ω resistor are connected in parallel across an ideal 9.0 V source. Use the loop rule to find each branch current and the junction rule to find the total current.

77 P-09-E-23, E-24 2+1 marks

A uniform potentiometer wire balances a 1.50 V standard cell at 60 cm. A second cell balances at 48 cm with the same wire current. Calculate the second cell’s e.m.f. and state the galvanometer reading at balance.

78 P-09-E-14, E-25 1+2 marks

An LDR is connected in series with a fixed resistor across a constant-voltage supply. The output is measured across the LDR. Predict how the output changes when the light brightens and explain using potential division.

Magnetism and electromagnetic induction

Q79–Q87 • 9 QUESTIONS

79 P-09-E-30, E-31 2+1 marks

A horizontal wire of length 0.40 m carries 5.0 A east in a uniform horizontal field of 0.30 T directed north. Calculate the force magnitude and use Fleming’s left-hand rule to give its direction.

80 P-09-E-32, E-33 1+2 marks

A proton and an electron enter the same uniform magnetic field with equal speeds and in the same direction, perpendicular to the field. Compare the magnitudes of the initial forces and explain why their paths bend in opposite directions.

81 P-09-E-33 1+2 marks

A charged particle enters a uniform magnetic field at right angles to it. Describe the path and explain why its speed stays constant even though its velocity changes.

82 P-09-E-34 2+1 marks

In a velocity selector, the electric and magnetic forces on a charged particle are opposite. If $E = 1.2 \times 10^4$ V/m and $B = 0.040$ T, calculate the speed for no deflection. State which force is larger for a faster particle entering in the same direction.

83 P-09-E-35, E-36 2+1 marks

Sketch the magnetic field of a current-carrying solenoid, showing the relative spacing of the field lines inside and outside. State the effect of inserting a soft iron core while keeping the current constant.

84 P-09-E-37 1+2 marks

Two long parallel wires carry currents in opposite directions. State whether they attract or repel and explain by considering the field of one wire and its force on the other.

85 P-09-E-38, E-39, E-40 2+1 marks

A 50-turn coil has an area of 0.020 m² per turn. A uniform field of 0.15 T is perpendicular to the coil’s plane. Find the flux through one turn and the total flux linkage.

86 P-09-E-41, E-42 2+1 marks

The magnetic flux through each turn of a 100-turn coil drops uniformly from 0.0060 Wb to 0.0020 Wb in 0.20 s. Calculate the magnitude of the induced e.m.f. and state whether the induced field reinforces or opposes the original field.

87 P-09-E-41 1+2 marks

The north pole of a bar magnet is pushed toward a closed conducting coil. Identify the pole formed on the near face of the coil. Predict what happens to the induced current when the magnet is held still, and explain.

Special relativity

Q88–Q92 • 5 QUESTIONS

88 P-11-F-01 1+2 marks

A bus moves straight at constant speed, then brakes. In which phase can its cabin be treated as an inertial frame? Explain the forward motion of a loose bag relative to the bus during braking.

89 P-11-F-02, F-04 1+2 marks

A spacecraft moving at 0.60c relative to Earth sends a light pulse forward. State the speed measured on the spacecraft and on Earth. Identify the postulate that justifies your answer.

90 P-11-F-05 1+2 marks

Two brief flashes at opposite ends of a railway platform are simultaneous in the platform frame. A train travels along the platform. Must the flashes also be simultaneous in the train frame? Explain using the relativity of simultaneity.

91 P-11-F-05 1+2 marks

A clock aboard a probe moving at 0.80c measures 6.0 μs between two flashes emitted beside it. Identify the proper time and calculate the interval measured in the Earth frame.

92 P-11-F-05 1+2 marks

A probe measures its own length as 40 m. It passes a laboratory at 0.60c with its length parallel to its velocity. Identify the proper length and calculate the length measured in the laboratory frame.

Nuclear and particle physics

Q93–Q100 • 8 QUESTIONS

93 P-11-F-07, F-08 2+1 marks

A nucleus with nucleon number 226 and proton number 88 emits an alpha particle. Find the daughter nucleus's nucleon and proton numbers. State the composition of the emitted particle.

94 P-11-F-10, F-11 1+2 marks

Electrons emitted in the same type of beta-minus decay have different kinetic energies. Name the other light particle emitted and explain how sharing the available energy accounts for the continuous spectrum.

95 P-11-F-13 2+1 marks

A proton has quark content *uud* and a neutron has *udd*. The up and down quark charges are $+\frac{2}{3}e$ and $-\frac{1}{3}e$. Calculate the net charge of each and explain how a neutron can contain charged particles and still be electrically neutral.

96 P-11-F-14, F-16 1+1+1 marks

Classify these three particles as a baryon, meson or lepton, giving a reason for each classification. The particles are a neutron, an up–antidown quark pair bound together, and an electron neutrino.

97 P-11-F-10, F-15 2+1 marks

In beta-plus decay, a proton changes into a neutron. Identify the quark change and the charged particle emitted. Name the accompanying neutral lepton.

98 P-11-F-17 1+1+1 marks

For each interaction, name an exchange particle. The interactions are electromagnetic repulsion between electrons, the strong interaction between quarks, and the weak interaction in beta decay.

99 P-11-F-18 1+2 marks

Name the particle associated with the Higgs field. Explain what it means for an elementary particle to acquire mass through interaction with that field, and give one example of an elementary particle that acquires mass this way.

100 P-11-F-09, F-28 1+2 marks

An electron and a positron annihilate with negligible initial kinetic energy. State how their masses and charges compare. Name a possible pair of products and explain what happens to the initial rest energy.

Source basis The supplied Grade XI Physics assessment framework determines topic eligibility. The two FBISE model papers and the 2026 first annual papers served as references for question style and difficulty.